

EXPERIMENT 4

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Subject Name: Computer Network

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Experiment 3 – Study the workings of Static Routing.

Aim : To study and implement **Static Routing** in **Cisco Packet Tracer** by configuring a network consisting of **three interconnected routers**, where each router is connected to a switch and end devices (PCs), and to verify successful communication between different networks through manual routing configuration.

Objective

- To understand the concept and working of **static routing** in a computer network.
- To configure IP addresses on PCs, switches (if required), and router interfaces in Cisco Packet Tracer.
- To manually configure static routes on routers for communication between multiple networks.
- To verify end-to-end connectivity between all networks using network testing tools such as the **ping** command.

Theory:

Static routing is a routing technique in which the network administrator manually configures the routing paths on routers. Unlike dynamic routing protocols, static routing does not automatically update routing tables when the network topology changes. Instead, each router is provided with explicit information about the destination networks and the corresponding next-hop IP addresses or exit interfaces

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Practical Steps:

Step 1: Create LAN 1

- Take **1 Switch, 2 PCs, and 1 Router (Router3)**.
- Connect both PCs to the switch using **Copper Straight-Through** cables.
- Connect the switch to **Router3 (GigabitEthernet 0/0)**.

Step 2: Create LAN 2

- Take **1 Switch, 2 PCs, and 1 Router (Router4)**.
- Connect both PCs to the switch.
- Connect the switch to **Router4 (GigabitEthernet 0/0)**.

Step 3: Create LAN 3

- Take **1 Switch, 2 PCs, and 1 Router (Router5)**.
- Connect both PCs to the switch.
- Connect the switch to **Router5 (GigabitEthernet 0/0)**.

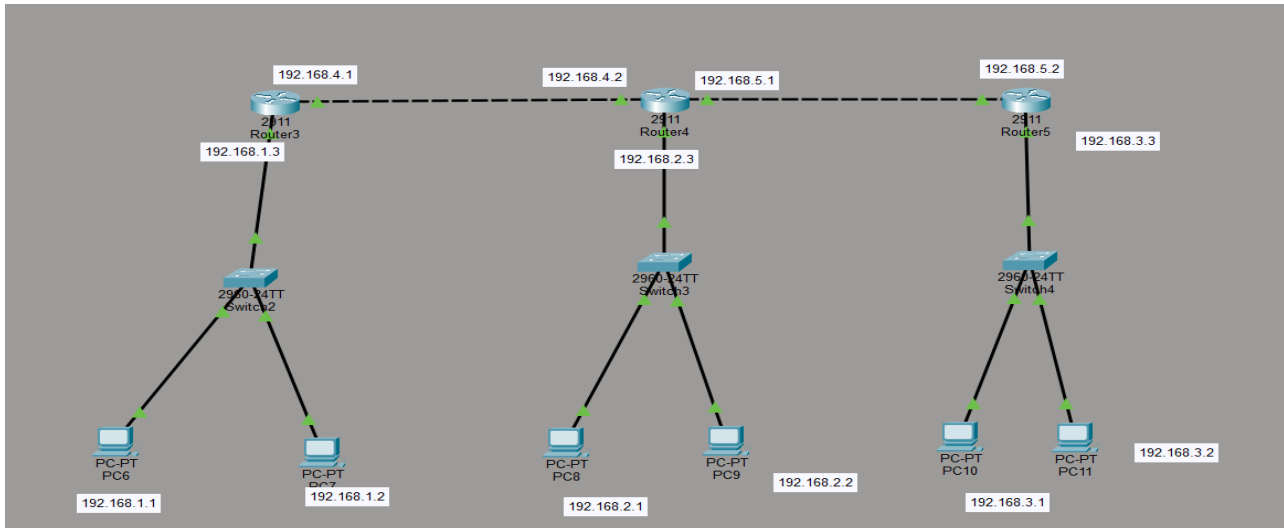
Step 4: Connect the Routers

- Connect **Router3** to **Router4** using a **Serial DCE/DTE cable**.
- Connect **Router4** to **Router5** using another **Serial DCE/DTE cable**.

Step 5: Connect the Routers

- Connect **Router3** to **Router4** using a **Serial DCE/DTE cable**.
- Connect **Router4** to **Router5** using another **Serial DCE/DTE cable**.

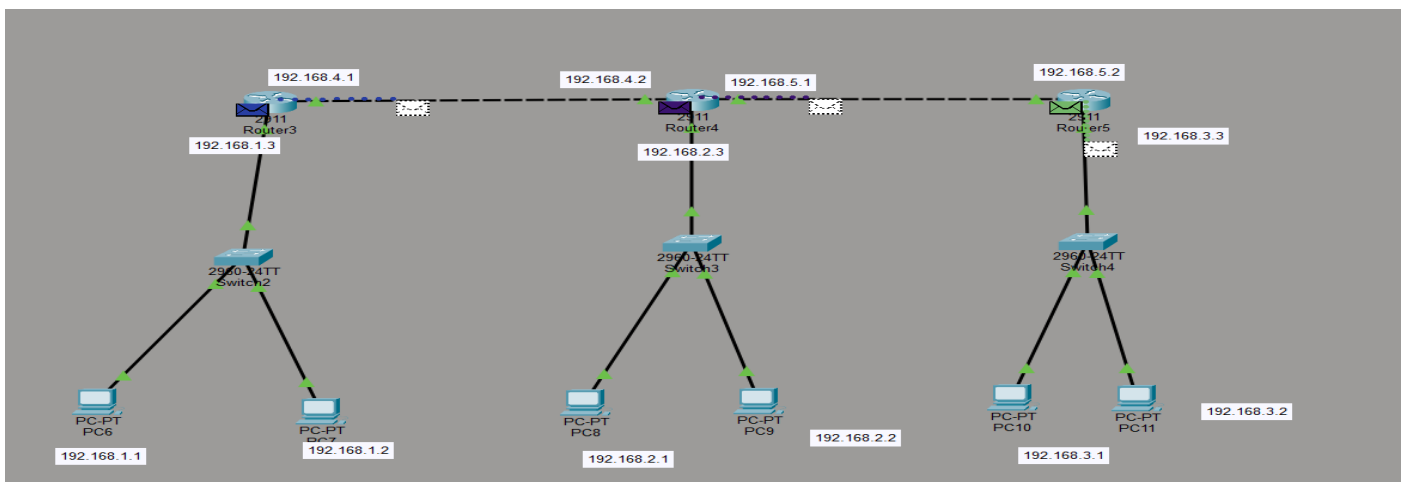
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Step 6: Configure Static Routing

Configure static routes on each router so that all three LANs can communicate.

- Give Network Address of Network 192.168.2.0, 192.168.3.0 and their subnet mask and net hop to Router 3
- Give Network Address of Network 192.168.1.0, 192.168.3.0 and their subnet mask and next hop to Router 4
- Give Network Address of Network 192.168.1.0, 192.168.2.0 and their subnet mask and next hop to Router 5
- Connect **Router4** to **Router5** using another **Serial DCE/DTE cable**.



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Learning Outcome

After completing this experiment, the learner will be able to:

1. **Understand the concept of static routing** and its role in enabling communication between multiple networks using manually configured routing tables.
2. **Develop practical skills in configuring network devices** by assigning IP addresses, subnet masks, default gateways, and router interfaces in Cisco Packet Tracer.
3. **Gain hands-on experience in configuring static routes** on multiple routers and establishing end-to-end connectivity between different LANs.
4. **Learn to verify and troubleshoot network communication** using tools such as the **ping** command, **Simple PDU**, and router diagnostic commands to ensure successful data transmission across interconnected network